Does the Habermas Machine under-represent minority opinions?

A replication of Figure 4C of Tessler et al. (2024)

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How to read this document. The prose is my notebook draft, transcribed with light edits. Red text fills a placeholder that the draft left open. Blue text replaces something the draft got wrong; a footnote gives the original wording and the source that corrects it. Every number is generated from `report/values.json` (see `report/build_parameters.py`); none is typed into this document. Code references are pinned to commit `ab58734c1`.

Introduction

In the original Habermas Machine (HM) paper, the authors consider the possibility that the HM might exhibit a "tyranny of the majority" phenomenon, in which minority perspectives are under-represented in the HM's output, relative to their prevalence among the population. They found that it did not do so. To do this, they developed a novel method for describing to what degree a summary statement reflected majority and minority opinions. This finding is significant because the HM's design goal was to produce a consensus statement that all participants would believe represented their opinion. In this report, I aim to replicate this finding using the publicly-released data from the original HM paper. To cut to the chase, I was unable to replicate this finding.

Approach to reproduction

Code: `hm_fig4c/data.py:97-187` build_tables() `hm_fig4c/data.py:221-232` prepare() `hm_fig4c/preprocess.py:50-61` preregistered_launches() `hm_fig4c/preprocess.py:41-47` filter_by_number_of_groups_of_min_size()

The authors released granular data from the experiments that they ran. They did not release the code that they used to perform the "tyranny of the majority" claim. I used details in their Supplementary Materials (SM) to attempt to replicate their analysis pipeline. To understand the deviations between the original analysis and my reproduction, I calculated several estimands reported in the original paper. There are several points where the SM are ambiguous and leave an aspect of the analysis pipeline unspecified. I approach this in the following manner: First, I present the analysis pipeline which I believe is most likely to match the original analysis. Then, after I have presented this "most-likely" pipeline, I perform an exhaustive sensitivity analysis, in which I vary every ambiguous detail (Sensitivity analysis).

Scope. This report replicates one claim: the minority-weight result shown as Figure 4C in the main paper.

What is the Habermas Machine (HM)?

Briefly, the HM is a tool for abstractive summarisation of people's opinions. A group of people each provide their opinion on a controversial topic. The HM iteratively produces and refines a consensus statement summarising the opinions in the group. First, it generates n initial statements. It then ranks these statements according to a simulated vote, in which it attempts to simulate the preferences of each opinion's author across the initial statements. The first-vote winner thus produced is shown to each (human) author. Each author provides feedback. This feedback is provided to the LLM. It generates a second round of candidate statements. These again go through a simulated vote, resulting in the final statement.

The finding that is the focus of this report is that when one compares the extent to which minority opinions are represented in the population (proportion of 0.29), this is roughly the same as how represented minority opinions are in the initial generated statements (0.28), and is less than how well represented minority opinions are in the final statement (0.36).[1] This is shown in Figure 1, which is copied from Figure 4(c) in the paper.

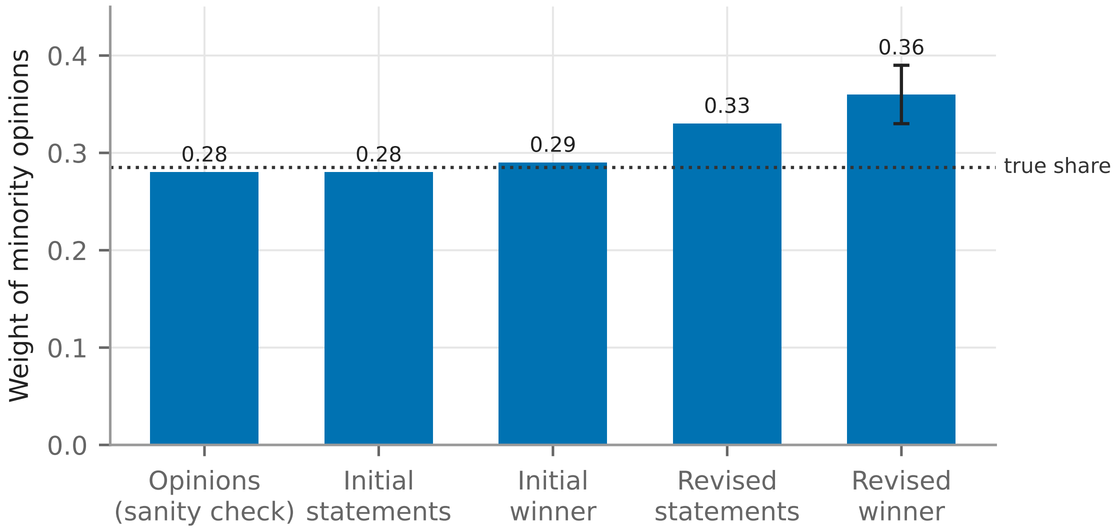

Figure 1: The paper's own result, redrawn from the data labels of SM Figure S60 ("Position Embedding" panel), which SM 5.2.2 identifies as the expanded version of main-text Figure 4C. Dotted line: the true proportion of minority opinions (0.285). Error bar on the revised winner: ±0.03, one standard error of the estimated regression coefficient, as printed in the paper. $n = 1047$ rounds.

What is a minority (or majority) group?

Code: `hm_fig4c/analysis.py:59-93` assign_minority() `hm_fig4c/pipeline.py:40-45` select_cohort()

In the papers' experiments, groups of between 4–5 people took part in the deliberative process mediated by the HM. Before writing their free-text opinion, they indicated their level of agreement with a political proposition using a Likert scale of 1–7. It was thus possible to provide a neutral (4) opinion. A group was defined as having a minority group if there was an unequal

[1] The dotted "true proportion" line in SM Figure S60 is at 0.285; the 0.28 and 0.29 bars are the initial candidates and the initial winner respectively. The comparison in this sentence is therefore between 0.285 and 0.36.

amount of agreeing (5–7) and disagreeing (1–3) opinions. Groups without a minority group were excluded from this analysis.

Reproduction parameters. The primary sample is the pre-registered groups of cohorts 1–3: 1047 rounds, of which 560 contain a minority under this rule, giving a true minority share of 0.262.[2]

How does the HM paper measure how well-represented minority opinions are?

This part of the analysis has two components. First, each participant's free-text opinion and the statements produced by the HM (including intermediate statements produced in its iterative process) are given a scalar score to measure to what extent they agree or disagree with the political proposition. I refer to this score as the "position component score".[3]

The position component scores of original human opinions and LLM-generated statements are then used to measure how similar the average position component scores among a minority group are, to the position component score of the LLM-generated statement.

I now detail my reproduction of each of these steps.

Measuring the position component score

Code: `hm_fig4c/data.py:38-47` endpoint_texts() `hm_fig4c/analysis.py:23-38` position_axis_scores() `hm_fig4c/analysis.py:41-55` score_texts() `hm_fig4c/embed.py:26-62` run()

To measure the position component score of a human-authored opinion or a LLM-generated statement, it is first mapped into a 768-dimension embedding, and then projected onto the vector going between two points.[4]

The main detail missing from the SM is what exact embedding model was used for this process. While the SM describes Sentence-T5, there are various sizes of this model that could be used.

[2] The paper reports 0.285. The gap is almost entirely the treatment of neutral raters. SM 5.4.1 counts a neutral (4) rating as non-minority: "we may define the 'minority' side of the Likert scale as the side that contains the smaller number of opinions, and define all other opinions as 'non-minority' opinions—which include 'majority' opinions [. . .] as well as 'neutral' opinions (i.e. with a Likert rating of 4), if any." Applied literally, with rounds that have equal numbers on each side excluded, that rule gives 0.262 on these rounds. Two other readings land on the paper's figure: dropping neutral raters from their groups before the sides are counted gives 0.289 (560 rounds), and keeping the tied rounds with a fixed side taken as the minority gives 0.284 (659 rounds). The treatment of neutral raters is carried through the sensitivity analysis below.

[3] The name follows SM 5.1.1, where the position component of a text embedding is its value along the position axis. It is best understood as a way of measuring the level of agreement with the proposition, similar to a political valence score: a single number that places a text between the negating and the affirming position on the question.

[4] SM 5.1.2 gives the method for constructing the two endpoints — a generic prefix ("Yes, I agree." or "No, I disagree.") concatenated with the question's own affirming or negating position statement — and one illustrative example, but not the exact phrases used for each question. Here each endpoint is the prefix followed by the position statement released with the data; the axis is the unit vector from the negating to the affirming endpoint (SM eq. 3) and the score is the projection onto it (SM eq. 5).

Validating the position component score

Code: `hm_fig4c/pipeline.py:54-57` fig4a_correlation() `report/compute_values.py:39-56` marginal_r2()

The original authors validate their position component score method by using the original human responses, comparing the human's self-reported Likert scores against the estimated position component scores derived from the method above. In the original paper, they report an R^2 of 0.41.[5] In our reproduction, the R^2 of each embedding model is shown in Table 3 (Appendix A).

Based on these results, we choose model sentence-t5-large as the most-likely model used in the original analysis. Its marginal r^2 of 0.392 is within 0.018 of the paper's 0.41, and its Pearson r of 0.633 against the paper's 0.64 says the same thing: the position axis this report is built on is about as good as the one the paper used. The other size that could be run here, sentence-t5-base, enters the sensitivity analysis as the alternative model.

Measuring representativeness

Code: `hm_fig4c/analysis.py:97-108` convex_lstsq() `hm_fig4c/analysis.py:111-132` convex_fit_with_se() `hm_fig4c/analysis.py:135-156` build_design() `hm_fig4c/analysis.py:172-185` minority_weight()

Given a set of human-authored opinions $\{x_i \mid i \in 1, \ldots, n\}$, a partition of opinions into minority M & non-minority N sets, position component scores $\{a_i \mid i \in 1, \ldots, n\}$, the "representation" of an opinion x_i is given by λ_i in

$$\text{set } \lambda_i \text{ to minimise } \Big(a_0 - \sum_i \lambda_i a_i\Big)^2 \qquad \text{such that} \qquad \sum_i \lambda_i = 1 \;\; \& \;\; \lambda_i \geq 0.$$

To calculate the representation of M & N, simply calculate the average representation of their elements: $\lambda_M := \frac{1}{|M|} \sum_{x_i \in M} \lambda_i$.

The implementation check

Code: `hm_fig4c/analysis.py:223-246` opinion_self_regression()

To test that the implementation of this procedure works as expected, the original authors ran a check where each human opinion in turn is treated as the target and regressed on convex combinations of that same set of opinions. Averaged over the set, the recovered minority weight should come back as simply the proportion of opinions in the minority.[6] We replicate this implementation check. It passes: the recovered weight is 0.263 against a true share of 0.262; the paper reports 0.28 against its true share of 0.285. This is the leftmost bar of Figure 2.

[5] The paper does not say exactly how this R^2 is computed. Appendix A sets out the candidate definitions, how I settled on one, the regression that produces it, and its value under each embedding model.

[6] Corrected. The draft read: "the output statement was simply one of the m human opinion inputs, and repeated this m times (once for each input). [this may be incorrect] After averaging across these runs, the average representation $\bar{\lambda}_i$ of an input x_i should be $\frac{1}{m}$." The draft's own flag was right. The paper's Materials and Methods describe it as: "As a sanity check, we similarly regress individual opinions on convex combinations of those same opinions, which we verify to simply yield the proportion of opinions in the minority." The target of the check is the *minority share*, not $1/m$ per input.

Results

Code: `hm_fig4c/pipeline.py:74-105` run_minority_analysis() `hm_fig4c/analysis.py:188-220` cluster_bootstrap() `report/figures.py:1-128` (whole file)

We fail to replicate the results from the original paper. Figure 2 displays an adjusted version of Figure 1 above, where we contrast the original paper's results against our own.[7]

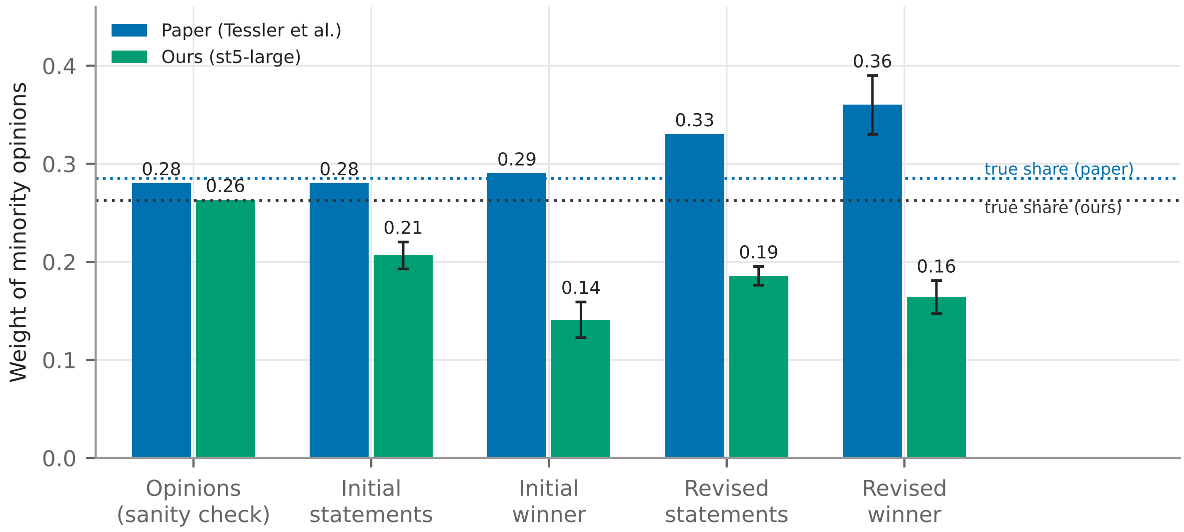

Figure 2: Minority weight at each phase: the paper against this reproduction under sentence-t5-large. Error bars are ±1 standard error of the estimated regression coefficients, as in SM Figure S60. The two dotted lines are the true minority share in each case (0.285 for the paper, 0.262 here; the difference is the treatment of neutral raters). $n = 560$ rounds with a minority, from 1047 pre-registered rounds in cohorts 1–3.

Three observations: 1) The sanity check reproduces exactly — the leftmost bars sit on their respective true shares — so the regression machinery is doing what the paper says it does. 2) Every statement phase, by contrast, sits below the true share rather than at or above it: the revised winner carries 0.164 ± 0.017 against the paper's 0.36. 3) The shape differs: in the paper the four phases rise monotonically, whereas here the winner-selection step *drops* minority weight at both rounds (initial candidates $0.206 \rightarrow$ initial winner 0.141), and the revision step partially restores it (0.164). The paper's rise from initial winner to revised winner is therefore present (+0.023, 95% CI $[-0.001, 0.044]$, $p = 0.030$), but it starts from a lower base and does not carry the final statement above proportionality; measured from the initial candidates instead, the change over the whole process is -0.043 (95% CI $[-0.077, -0.008]$).

Validation checks

Code: `hm_fig4c/pipeline.py:60-71` fig4b_within_range()

[7] Error bars in both figures are ±1 standard error of the estimated regression coefficients, which is how SM Figure S60 defines them. The SM does not say how that standard error is obtained for a constrained regression. Here the constrained fit is treated as an ordinary least-squares fit on its active set (the coefficients above zero) with the sum-to-one constraint eliminated; the standard error of the summed minority coefficients is read from the resulting covariance matrix, and the level-specific standard errors are combined with the same round-count weights as the estimate itself (SM eq. 19). As a check on that construction, a cluster bootstrap over rounds gives a standard error of 0.017 for the revised winner against the analytic 0.017.

Everything above is only interesting if the pipeline is sound, so the checks that bear on that are collected in Table 1. Three of the four independent checks land on the paper's values, which is the basis for reporting the disagreement rather than assuming a bug. The one that does not — the share of statements falling inside the range of their group's opinions — is lower here than in the paper, and is the clearest remaining candidate for an unmodelled difference.

Table 1: Checks on the reproduction, independent of the minority-weight result.

check	paper	ours (sentence-t5-large)
Sanity check: opinions regressed on themselves recover the minority share	0.28 vs 0.285	0.263 vs 0.262
Position axis quality: marginal r^2 of Likert on position component score	0.41	0.392
Fig. 4A: correlation of position component score with Likert rating	0.64	0.63
Fig. 4B: statements falling within the range of their group's opinions	0.96	0.86

Sensitivity analysis

Code: `hm_fig4c/pipeline.py:114-126` sensitivity_grid() `hm_fig4c/pipeline.py:129-139` run_sensitivity() `report/compute_values.py:81-94` sensitivity_summary()

As noted above, there are several aspects of the analysis plan which are not specified in the SM. These are:

Table 2: Underspecified choices, the options considered, and the option taken in the primary specification above.

ambiguity	possible options	option chosen above
ST5 model size	base, large, xlarge, xxl	large (Appendix A)
Neutral (Likert 4) raters	count as non-minority; drop the rater; drop the whole group	count as non-minority (SM 5.4.1)
Basis for splitting minority / non-minority	Likert rating; sign of the position component score (SM Fig. S62)	Likert rating (SM Fig. S60)
Column order: how a round's opinions are assigned to the regressors	order in the released data; sorted by position component score; random	data order

The last row needs a word of explanation. The regression at each level of division fits one coefficient per column of the design matrix, and each row of that matrix is one round's opinion scores, so a column pools "the j-th opinion" across rounds. SM eq. 13 indexes a round's opinions by j without saying how j is assigned. Here the minority opinions always occupy the first columns, and the three options above fix the order within the minority block and within the non-minority block. The minority weight sums the minority columns, which removes the dependence on order within the minority block when the minority has one member, but the order of the non-minority columns still changes the fit.

I reran the analysis with every combination of these decisions (24 runs).[8] Across the runs, I

[8]Two model sizes × three neutral-rater treatments × three column orders when the split is by Likert rating, plus two model sizes × three column orders when the split is by the sign of the score, where there are no

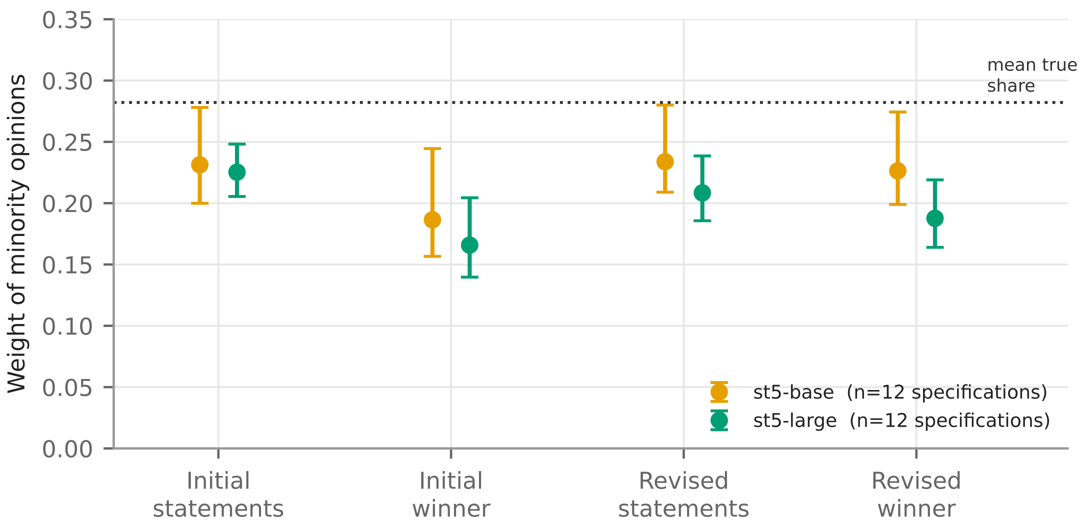

Figure 3: Mean minority weight at each phase across the 12 specifications run under each embedding model; whiskers span the full range (minimum to maximum) of those specifications. Dotted line: the mean true minority share across all 24 runs (0.282). No specification places the revised winner above its own true share.

calculated the following metrics:

- Proportion of runs where minority representation increases from initial to final: 12%
- Proportion of runs where minority representation is larger in final than the prevalence among population (0.262):[9] 0%. The largest revised-winner weight in the sweep is 0.274, against a mean true share of 0.282.
- Average representation at each of the 4 stages, with the range across runs: shown in Figure 3.

Discussion

If the above reproduction of the analysis pipeline is correct, then it would appear that the HM actually suppresses minority opinions, if we follow the original paper's interpretation of the above measurement method.

Notably, we are able to reproduce several parameter estimates from the paper, including the validation of the embedding-based position component score and the implementation check of the regression method. We fail to replicate the substantive findings for this part of the paper, however.

This finding actually aligns somewhat with the finding that people's beliefs shifted towards the majority position (although there are many possible explanations for this finding).

neutral raters: 24 runs, 12 per model. The xlarge and xxl sizes could not be run here (Appendix A).

[9]Corrected. The draft gave 0.291. That is the true share under one variant — neutral raters dropped — not under the primary specification, where it is 0.262. Each run is compared against its own true share rather than a single fixed number, since the minority rule changes the share as well as the weight.

Appendix A: the paper's R^2 statistic

Code: `hm_fig4c/pipeline.py:54-57` fig4a_correlation() `report/compute_values.py:39-56` marginal_r2()

What the paper says. The main text reports: "A random-effects estimator of the regression coefficient for Likert ratings on position embeddings yields a marginal r-squared value of 0.41 (SM 5.1.3)." SM 5.1.3 adds that a model taking the top ten residual embedding components as additional inputs has marginal and conditional r^2 of 0.41 and 0.45, "similar to the model containing only the position component". That leaves three things unspecified: which R^2 is meant — the marginal R^2 of a mixed model, or the plain Pearson r^2 of the scatter in Figure 4A, whose $r = 0.64$ squares to 0.41; what the random effect is grouped by (question, round or participant); and whether only the intercept or also the slope varies by group.

How I settled on a definition. "Marginal" and "conditional" r^2 are the standard names for the Nakagawa–Schielzeth decomposition of variance explained in a mixed model, and the main text names a random-effects estimator, so I take the statistic to be the marginal R^2 of a random-intercept model. I group by round, the unit at which a group forms and a question is answered, and let only the intercept vary, since a random slope would add a further unstated choice. Because the paper's Pearson r^2 and its marginal R^2 coincide at 0.41, the paper's own number cannot separate the two definitions, so Table 3 reports both. Under either definition the ranking of the embedding models, and with it the choice of sentence-t5-large, is the same.

The regression. For participant j in round i, with Likert rating y_{ij} and position component score s_{ij},

$$y_{ij} = \alpha + \beta\, s_{ij} + u_i + \varepsilon_{ij}, \qquad u_i \sim \mathcal{N}(0, \sigma_u^2), \quad \varepsilon_{ij} \sim \mathcal{N}(0, \sigma_\varepsilon^2),$$

fitted by restricted maximum likelihood on the 4979 opinions of the 1047 pre-registered rounds of cohorts 1–3. Writing σ_f^2 for the variance of the fixed-effect predictions $\hat{\beta} s_{ij}$,

$$R^2_{\text{marginal}} = \frac{\sigma_f^2}{\sigma_f^2 + \sigma_u^2 + \sigma_\varepsilon^2}, \qquad R^2_{\text{conditional}} = \frac{\sigma_f^2 + \sigma_u^2}{\sigma_f^2 + \sigma_u^2 + \sigma_\varepsilon^2}.$$

Under sentence-t5-large the fitted slope is $\hat{\beta} = 10.686$ (SE 0.204); under sentence-t5-base it is 11.042 (SE 0.259).

Table 3: Fit of the position component score to the Likert rating, by embedding model: marginal and conditional r^2 of the random-intercept model above, and the Pearson correlation of the raw scatter (the main-text Figure 4A quantity) with its square.

	marginal r^2	conditional r^2	Pearson r	Pearson r^2
sentence-t5-base	0.291	0.443	0.561	0.315
sentence-t5-large	0.392	0.505	0.633	0.400
sentence-t5-xlarge		not evaluated — see note		
paper	0.41	0.45	0.64	0.41

The xlarge row is empty for a hardware reason rather than a scientific one: sentence-t5-xl is a 3B-parameter model and this machine has no GPU and 7 GB of RAM, so it could not be embedded here. The two sizes that could be run bracket the question well enough to answer it — base falls well short of the paper's 0.41 and large lands essentially on it — and the direction of the remaining risk is worth stating plainly: if the authors used a larger ST5 than large, the axis would be slightly *better* than mine, and a better axis moves the minority weight further from the paper's value, not closer (Sensitivity analysis).